From Restaurant Growth to an Exact Partition Atlas: Ewens Laws, Block-Count Fluctuations, and Poisson Limits

HCS-C353 source-local reconstruction

3 September 2026

Abstract

We carry the one-parameter Chinese-restaurant insertion rule through a complete finite and asymptotic partition atlas. A labelled-partition induction proves exchangeability and the Ewens sampling formula; exact counting gives the full occupancy-vector law. The total number of blocks is factored into independent Bernoulli innovations, yielding its exact probability generating polynomial, almost-sure logarithmic law, and variance-normalized central limit theorem. A separate mixed falling-factorial identity proves joint independent-Poisson limits for every fixed collection of block sizes. A 5,238-row exact certificate checks finite coefficients and conventions but does not replace the asymptotic proofs. No priority or arithmetic-target claim is made.

Revision certificate. Exchangeability, EPPF, and exact occupancy-law closure.

1 Growth rule and complete theorem

Fix $\theta > 0$. Set $\Pi_1 = \{\{1\}\}$. Given a partition Π_n of $[n] = \{1, \dots, n\}$, customer $n+1$ starts a singleton block with probability $\theta/(\theta + n)$, or joins an existing block B with probability $|B|/(\theta + n)$. Let $K_n = |\Pi_n|$ and let $C_j(n)$ be the number of blocks of size j. Write $(a)^{\overline{n}} = a(a+1)\cdots(a+n-1)$ and use $(x)_{\underline{r}} = x(x-1)\cdots(x-r+1)$.

Theorem 1 (Finite-to-asymptotic partition atlas)**.** *For the preceding process:*

(i) *Π_n is exchangeable. A particular labelled partition $\{B_1, \dots, B_k\}$ has probability*

$$\mathbb{P}(\Pi_n = \{B_1, \dots, B_k\}) = \frac{\theta^k}{(\theta)^{\overline{n}}} \prod_{i=1}^{k} (|B_i| - 1)!. \tag{1}$$

(ii) *If $c_1, \dots, c_n \geq 0$ and $\sum_j j c_j = n$, then*

$$\mathbb{P}(C_j(n) = c_j, 1 \leq j \leq n) = \frac{n!}{(\theta)^{\overline{n}}} \prod_{j=1}^{n} \frac{(\theta/j)^{c_j}}{c_j!}. \tag{2}$$

(iii) *There are independent Bernoulli variables I_i with $\mathbb{P}(I_i = 1) = \theta/(\theta + i - 1)$ such that $K_n = \sum_{i=1}^{n} I_i$. Consequently*

$$\mathbb{E} z^{K_n} = \frac{(\theta z)^{\overline{n}}}{(\theta)^{\overline{n}}}. \tag{3}$$

(iv) *As $n \to \infty$,*

$$\frac{K_n}{\log n} \longrightarrow \theta \quad \textit{almost surely}, \qquad \frac{K_n - \mathbb{E}K_n}{\sqrt{\operatorname{Var} K_n}} \Longrightarrow N(0,1). \tag{4}$$

(v) *For every fixed m,*

$$(C_1(n), \dots, C_m(n)) \Longrightarrow (Z_1, \dots, Z_m), \qquad Z_j \ \textit{independent}, \quad Z_j \sim \operatorname{Poisson}(\theta/j). \tag{5}$$

2 Labelled induction and the occupancy law

Formula (1) holds at $n = 1$. Assume it at n. If $n+1$ creates a new block, multiplication by $\theta/(\theta + n)$ adds one power of θ and the next rising-factorial denominator. If it joins a block of size b, multiplication by $b/(\theta + n)$ changes $(b-1)!$ to $b!$. Every labelled partition at time $n+1$ has a unique predecessor obtained by deleting label $n+1$, so induction proves (1) without multiplicity ambiguity. Its dependence only on block sizes proves exchangeability.

For a count vector $c = (c_1, \dots, c_n)$, the number of labelled set partitions with those counts is

$$\frac{n!}{\prod_{j=1}^{n} (j!)^{c_j} c_j!}. \tag{6}$$

Multiplying (6) by (1), with $\prod_i (|B_i| - 1)! = \prod_j ((j-1)!)^{c_j}$, gives (2). Summing (2) also yields

$$(\theta)^{\overline{n}} = \sum_{k=1}^{n} c(n,k)\theta^k,$$

where $c(n,k)$ is the unsigned Stirling number of the first kind.